Wheat - Yellow rust (stripe rust)

RAG record ADV-010 | canonical key: wheat_yellow_rust

Record ID	ADV-010
Crop / crop key	Wheat / wheat
Condition / disease key	Yellow rust (stripe rust) / yellow_rust
Main pathogen	Puccinia striiformis f. sp. tritici
RAG aliases	wheat yellow rust, stripe rust, Puccinia striiformis, yellow stripes wheat

Farmer quick guide: treatments protect healthy plant tissue and slow disease spread. Dead or badly damaged leaf tissue will not become green again.

How to recognize it

- Bright yellow powdery pustules form clear stripes that run along the length of the leaf.
- If you rub an active pustule, yellow spores may come off on your fingers.

Weather or field conditions that favour it

Cool weather, dew, fog and intermittent rain favour yellow rust, especially in northern and hill wheat zones.

What to do now

- Inspect the field often during cool cloudy weather. Find and treat early foci before they spread widely.
- Grow locally recommended resistant varieties and destroy volunteer wheat that can carry rust between seasons.
- Use a registered fungicide as soon as yellow rust is confirmed and local thresholds/advisories indicate treatment.

Chemical treatment - only after the disease is confirmed

Chemical / formulation	Rate from cited source	How to use this information	Source
Propiconazole 25% EC	500 ml/ha in about 750 L water	Government PPQS yellow-rust IPM package	[S1]

Chemical / formulation	Rate from cited source	How to use this information	Source
		lists this rate; waiting period stated as 30 days.	
Propiconazole 25% EC	0.1% (1 ml/L) in ICAR field advisory	ICAR/DWR has also advised this concentration when yellow rust is noticed. Follow the current product label where application-volume instructions differ.	[S2]

Safety and compliance: Use only a product whose current Indian label allows the crop and target disease. Follow the label rate, water volume, re-entry interval and pre-harvest interval. Wear label-required PPE. Do not mix products unless the label or a qualified extension adviser permits it.

Prevention and integrated management

- Use resistant varieties recommended for the wheat zone.
- Avoid excessive nitrogen.
- Avoid waterlogging.
- Destroy volunteer wheat and maintain field sanitation.

RAG guardrail

The AI should treat the chemical table as source-grounded examples, not permission to invent a different pesticide or dose. If the retrieved source does not match the crop, disease and formulation, return non-chemical advice and request local verification.

Sources used for this record

[S1] Directorate of Plant Protection, Quarantine & Storage (PPQS), Government of India. IPM Package of Practices for Management of Yellow Rust of Wheat. https://ppqs.gov.in/sites/default/files/pop_for_management_of_yellow_rust_of_wheat.pdf

[S2] Indian Council of Agricultural Research / Directorate of Wheat Research. Stripe Rust (Yellow Rust) of Wheat control advisory. https://icar.gov.in/en/node/5291